An Exact Reachability Criterion for Digit-Sum–Bounded Lattice Walks

Adam Benzan*

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Abstract

Fix a base $b \geq 2$ and a threshold N, and call a lattice cell $(x, y) \in \mathbb{Z}^2$ *allowed* if $\mathsf{s}(|x|)+\mathsf{s}(|y|) \leq N$, where s denotes the base-b digit sum. We characterize the set of cells reachable from the origin by 4-connected paths through allowed cells: (x, y) is reachable if and only if every proper digit-prefix pair of $(|x|, |y|)$ satisfies an explicit budget inequality with exactly one unit of slack, together with allowedness of the cell itself. The proof is elementary and constructive. Corollaries: reachability coincides with monotone (staircase) reachability; the reachable-cell count is computed exactly by a one-dimensional walk recursion in time polynomial in $\log N$-many digits; and the asymptotic density of the reachable set within the allowed set is governed by classical one-boundary fluctuation theory.

1 Setting and statement

Fix a base $b \geq 2$ and write $B = b - 1$. Fix a threshold $N \geq 1$ and let $k = \lfloor N/B \rfloor + 1$, so that

$$Bk > N \;\geq\; B(k-1). \tag{1}$$

For an integer $v \geq 0$ let $\mathsf{s}(v)$ be its base-b digit sum. A cell $(x, y) \in \mathbb{Z}^2$ is *allowed* if $\mathsf{s}(|x|) + \mathsf{s}(|y|) \leq N$, and *reachable* if there is a path from the origin to (x, y), moving one unit at a time in the four axial directions, all of whose cells are allowed. (The origin is allowed.)

For $0 \leq x < b^k$ write the digits of x most significant first, $x = x_1 x_2 \cdots x_k$ (leading zeros permitted), and for $0 \leq s \leq k$ let $x\langle s\rangle$ denote the integer with digits $x_1 \cdots x_s$. For a pair (x, y) in the quadrant define the *prefix sums*

$$P_s \;=\; \mathsf{s}(x\langle s\rangle) + \mathsf{s}(y\langle s\rangle), \qquad s = 0, 1, \ldots, k.$$

Theorem 1 (Reachability criterion)**.** *Let $0 \leq x, y < b^k$. Then (x, y) is reachable if and only if*

(i) $P_s \;\leq\; N - B(k-s) + 1$ *for every s with $1 \leq s < k$, and*

(ii) $P_k = \mathsf{s}(x) + \mathsf{s}(y) \;\leq\; N$.

Moreover every reachable cell of the closed quadrant satisfies $x, y < b^k$, and the full-plane reachable set is the image of the quadrant one under the four reflections $(x, y) \mapsto (\pm x, \pm y)$.

*Joint exploration with Claude (Anthropic). All results verified computationally against exhaustive search; see the repository scripts.

The empty prefix is excluded from (i): $P_0 = 0$, while the bound $N - Bk + 1$ may be negative by (1). Equivalently, in walk form: put $d_0 = Bk - N$ and $d_s = d_0 + \sum_{t \le s}(x_t + y_t - B)$. Then (i)–(ii) read

$$d_s \le 1 \ (1 \le s < k), \qquad d_k \le 0.$$

We use three elementary facts about digit sums.

Lemma 2. *For all integers $X \ge 1$ and $m \ge 0$:*

(D1) $\mathsf{s}(X) \le \mathsf{s}(X-1) + 1$*; more precisely* $\mathsf{s}(X) = \mathsf{s}(X-1) + 1 - Bt$ *when X ends in exactly t zeros.*

(D2) $\mathsf{s}(X \cdot b^m - 1) = \mathsf{s}(X-1) + Bm$.

(D3) *for $m \ge 1$ and $1 \le a < b$:* $\max\{\mathsf{s}(t) : 0 \le t \le a \cdot b^m\} = (a-1) + Bm$*, attained at $t = a \cdot b^m - 1$.*

Proof. (D1): incrementing either increases the last digit by one, or turns a digit-string suffix $d\,B \cdots B$ (t trailing copies of B) into $(d{+}1)\,0 \cdots 0$. (D2): $Xb^m - 1 = (X-1)b^m + (b^m - 1)$, whose digits are those of $X - 1$ followed by m copies of B. (D3): write $t = a'b^m + r$ with $0 \le r < b^m$. If $t < ab^m$ then $a' \le a - 1$ (a single digit) and $\mathsf{s}(t) = a' + \mathsf{s}(r) \le (a-1) + Bm$, with equality at $a' = a - 1$, $r = b^m - 1$; the endpoint $t = ab^m$ has $\mathsf{s}(t) = a \le (a-1) + Bm$ since $Bm \ge 1$. □

2 Reduction to the quadrant

Proposition 3. *The folding map $(x, y) \mapsto (|x|, |y|)$ sends allowed cells to allowed cells and neighbors to neighbors or equal cells. Consequently the reachable set of the plane is the fourfold reflection of the reachable set of the quadrant, and it suffices to prove Theorem 1 for paths confined to the quadrant.*

Proof. Allowedness depends only on $(|x|, |y|)$. If c, c' are neighbors then their images differ by at most one unit in one coordinate. □

Proposition 4. *Every reachable cell of the quadrant has $x, y < b^k$.*

Proof. Let w be the least integer with $\mathsf{s}(w) = N + 1$; explicitly $w = (s+1)b^q - 1$ where $N + 1 = Bq + s$, $1 \le s \le B$, so $w < b^k$. Every cell of the column $x = w$ has $\mathsf{s}(x) + \mathsf{s}(y) \ge N + 1$ and is forbidden; a quadrant path from the origin to any $x > w$ would have to occupy that column. Hence reachable cells have $x \le w - 1 < b^k$, and likewise for y. □

3 Necessity: wall certificates

Lemma 5 (Necessity)**.** *If (x, y) is reachable then* (i) *and* (ii) *hold.*

Proof. (ii) is allowedness of the endpoint. For (i), fix $1 \le s < k$ and let $m = k - s$. Partition the quadrant into *tanks* of scale m: tank (A, C) is the square $[Ab^m, (A{+}1)b^m) \times [Cb^m, (C{+}1)b^m)$. Let $T = (X, Y)$ with $X = x\langle s\rangle$, $Y = y\langle s\rangle$ be the tank containing (x, y).

If $(X, Y) = (0, 0)$ then $P_s = 0 \le N - Bm + 1$ by (1). Otherwise the origin lies outside T, so a reaching path π has a first step entering T: consecutive cells $c \notin T$, $c' \in T$. Note that any cell (u, v) with v in tank row Y has $\mathsf{s}(v) \ge \mathsf{s}(Y)$, since the digits of v are those of Y followed by a suffix. Four cases.

From the left (c in tank $(X{-}1, Y)$, so $X \geq 1$): then $c = (Xb^m - 1,\ y^*)$ with y^* in tank row Y. By (D2), $\mathsf{s}(Xb^m - 1) = \mathsf{s}(X{-}1) + Bm$, so allowedness of c gives $\mathsf{s}(X{-}1) + \mathsf{s}(Y) \leq N - Bm$, and (D1) yields

$$P_s = \mathsf{s}(X) + \mathsf{s}(Y) \ \leq\ \mathsf{s}(X-1) + 1 + \mathsf{s}(Y) \ \leq\ N - Bm + 1.$$

From below: symmetric, using the top (all-B) wall row of tank $(X, Y{-}1)$: $P_s \leq N - Bm + 1$.

From the right (c in tank $(X{+}1, Y)$): the *entered* cell lies in the rightmost column of T itself: $c' = ((X{+}1)b^m - 1,\ y^*)$, whose x-digit sum is $\mathsf{s}(X) + Bm$ by (D2). Allowedness of c' gives $P_s \leq N - Bm$ directly.

From above: symmetric: $P_s \leq N - Bm$.

In all cases (i) holds at s. □

Remark 6. The unit of slack in (i) is precisely the borrow in (D1): entries from the left or below certify the *neighbor's* wall, which is one digit cheaper than the target tank's own.

4 Sufficiency: the corner staircase

Lemma 7 (Sufficiency)**.** *For every threshold $M \geq 0$ and every $j \geq 1$: if $0 \leq x, y < b^j$ satisfy* (i)–(ii) *with (j, M) in place of (k, N), then there is a monotone staircase (unit steps $+x$ and $+y$ only) of allowed cells from $(0, 0)$ to (x, y).*

Proof. Induction on j. For $j = 1$, (ii) reads $x + y \leq M$ and the path $(0,0) \to (x,0) \to (x,y)$ has all digit sums $\leq x + y$.

Let $j \geq 2$, and let $a = x_1$, $c = y_1$ be the leading digits, x', y' the $(j{-}1)$-digit remainders. Condition (i) at $s = 1$ reads

$$a + c \ \leq\ M - B(j-1) + 1. \tag{$\star$}$$

Leg 1: to the corner of tank (a, c). Take the L-path $(0,0) \to (ab^{j-1}, 0) \to (ab^{j-1}, c\,b^{j-1})$. On the axis run, by (D3) every cell $(t, 0)$ with $t \leq ab^{j-1}$ has digit sum at most $(a-1) + B(j-1) \leq M$ by ($\star$) (using $c \geq 0$). On the column climb, every cell (ab^{j-1}, u) with $u \leq c\,b^{j-1}$ has digit sum at most $a + (c-1) + B(j-1) \leq M$, which is exactly ($\star$). (If $a = 0$ or $c = 0$ the corresponding leg is empty and the bounds only improve.)

Leg 2: inside the tank. The tank (a, c), entered at its bottom-left corner, is a copy of the same problem with $j - 1$ digit positions and threshold $M' = M - a - c$: a local cell (u, v) is allowed iff $\mathsf{s}(u) + \mathsf{s}(v) \leq M'$. For $2 \leq s \leq j$ put $s' = s - 1$; then

$$P_s \leq M - B(j-s) + 1 \iff P'_{s'} \leq M' - B\big((j{-}1) - s'\big) + 1,$$

where P' are the prefix sums of (x', y'), and (ii) becomes $\mathsf{s}(x') + \mathsf{s}(y') \leq M'$. Thus (x', y') satisfies the $(j{-}1)$-digit criterion at threshold M', and the induction hypothesis provides a monotone allowed staircase from the corner to it; translating by $(ab^{j-1}, c\,b^{j-1})$ adds exactly $a + c$ to every digit sum, preserving global allowedness.

Concatenating the legs completes the induction. □

Proof of Theorem 1. Propositions 3 and 4 reduce to the quadrant with $x, y < b^k$; Lemma 5 gives necessity; Lemma 7 with $(j, M) = (k, N)$ gives sufficiency (a staircase is in particular a path). □

Remark 8 (Sharpness). Necessity produces the bound $N - Bm + 1$ via the borrow in (D1); sufficiency consumes the same bound in the column climb of the L-path, whose worst cell $a\,b^{j-1} + (c\,b^{j-1} - 1)$ has digit sum $a + (c-1) + B(j-1)$. The two units of slack coincide, which is why the criterion is exact.

5 Corollaries

Corollary 9 (Monotone-staircase lemma)**.** *A cell is reachable iff it is reachable by a monotone staircase. In particular undirected reachability, directed (right/up) reachability, and the criterion coincide.*

Corollary 10 (Exact counting)**.** *Let* $w(\sigma) = \#\{(i,j) \in [0,B]^2 : i + j - B = \sigma\}$ *be the tent weights. Then the number* $Q(N)$ *of reachable quadrant cells equals the total weight of length-*k *walks* $d_0 \to d_k$ *with steps weighted by* w*, barrier* $d_s \le 1$ *for* $1 \le s < k$*, and endpoint* $d_k \le 0$*, where* $d_0 = Bk - N$*. This is a one-dimensional dynamic program over* $O(Bk)$ *states per step; the full-plane count is* $f(N) = 4\,Q(N) - 4\,w_0 + 1$ *with* w_0 *the least integer of digit sum* $N + 1$*.*

Corollary 11 (Asymptotics)**.** *On the peak line* $N \equiv B - 1 \pmod B$ *one has* $d_0 = 1$*, and* $Q(N)\,/\,b^{2k}$ *is the probability that the reflected tent walk stays nonnegative for* k *steps and ends* ≥ 1*. By the Sparre Andersen identity and the symmetry* $\Pr(S_n \ge 0) = \big(1 + p_n\big)/2$ *with* $p_n = \Pr(S_n = 0)$*, the reachable fraction of the allowed set decays as* $c_b/\sqrt{N}$ *with*

$$c_b \;=\; \frac{2\sqrt{b-1}}{\sqrt{\pi}}\,\exp\Big(\tfrac{1}{2}\sum_{n\ge 1}\frac{p_n}{n}\Big) \;=\; \frac{2\sqrt{b-1}}{\sqrt{\pi}}\,\exp\Big(-\frac{1}{4\pi}\int_{-\pi}^{\pi}\log\big(1-\varphi_b(\theta)\big)\,d\theta\Big),$$

where $\varphi_b(\theta) = \big(\sin(b\theta/2)/(b\sin(\theta/2))\big)^2$ *is the tent characteristic function — the Wiener–Hopf factorization constant of the tent kernel. In base* 2*,* $\sum_n p_n/n = \sum_n \binom{2n}{n}4^{-n}/n = 2\log 2$ *gives* $c_2 = 4/\sqrt{\pi}$ *in closed form; in base* 10*,* $c_{10} = 3.847174895\ldots$*, and* $c_3 = 2.478093$*,* $c_4 = 2.705016$*,* $c_5 = 2.921639$*, each matching independent walk-DP extrapolations.*

Corollary 12 (Base 2)**.** *For* $b = 2$*,* $Q(N) = \binom{2N+3}{N}$*. Moreover the reachable set is equinumerous with the set of allowed cells of digit-sum weight exactly* N *or* $N - 1$*, via an explicit bijection.*

Proof. Put $k = N + 1$ and interleave the bits of (x, y) into $w \in \{0,1\}^{2k}$, read as a ± 1 path (1 = up). The criterion says the path is ≤ 0 at every even time and ends ≤ -2. Even-time positions are even integers, so the first condition is equivalent to the uniform barrier "never reach +2". Reflecting at level 2 pairs the paths with max ≥ 2 ending at $-2m$ with free paths ending at $2m + 4$, whence

$$Q(N) = \sum_{m\ge 1}\Big[\tbinom{2k}{k-m} - \tbinom{2k}{k+m+2}\Big] = \tbinom{2k}{k-1} + \tbinom{2k}{k-2} = \tbinom{2k+1}{k-1} = \tbinom{2N+3}{N},$$

the middle equality by the symmetry $\binom{2k}{j} = \binom{2k}{2k-j}$ and the last by Pascal. The two surviving terms count the free words of weight $k - 1$ and $k - 2$ — the top two layers — and the Garsia–Milne chase of the reflection collapses to an explicit *surfacing map*: while w has $\le k - 3$ ones, complement the prefix up to and including the first hitting time of level -2 (each round adds two ones); the inverse complements prefixes up to first hits of +2. Bijectivity was verified exhaustively for $k \le 7$. □

Remark 13. The collapse is a parity accident of binary: base-2 digits give two-valued ± 1 half-steps, so even-time positions have fixed parity and the checkpointed condition becomes a uniform barrier. For $b \ge 3$ the half-steps take b values, no parity constraint survives, and no binomial closed form exists — consistent with c_b of Corollary 11 being a genuinely transcendental-looking factorization constant.

Remark 14. All statements were additionally verified computationally: the criterion against exhaustive breadth-first search for bases $3, 5, 10$ over their feasible ranges, and Corollary 10 against independent solvers up to $N = 989$ in base 10 (integer equality).